THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The midpoint of the segment joining $A(-3,\ 5)$ and $B(7,\ -1)$ is: 1

(a) $(2,\ 2)$ **(b)** $(2,\ -2)$ **(c)** $(5,\ -3)$ **(d)** $(4,\ 4)$

2. The degree of the polynomial $4z - 3$ is: 1

(a) 3 **(b)** 2 **(c)** 0 **(d)** 1

3. The number $\frac{22}{7}$ is: 1

(a) rational **(b)** irrational

(c) neither rational nor irrational **(d)** not a real number

4. Factorise $x^2 - 5x - 36$: 1

(a) $(x - 9)(x + 4)$ **(b)** $(x + 4)(x + 9)$ **(c)** $(x - 4)(x + 9)$ **(d)** $(x - 9)(x - 4)$

5. In a circle of radius 10 cm, a chord is 6 cm from the centre. The length of the chord is: 1

(a) 14 cm **(b)** 16 cm **(c)** 8 cm **(d)** 18 cm

6. The length of an arc subtending 90° at the centre of a circle of radius 21 cm ($\pi = \frac{22}{7}$) is: 1

(a) 33 cm **(b)** 66 cm **(c)** 16.5 cm **(d)** 132 cm

7. A bag contains 10 red, 6 blue and 4 green balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{3}{10}$

8. The 6th term of the GP with first term 3 and common ratio 2 is: 1

(a) 95 **(b)** 192 **(c)** 96 **(d)** 97

9. Which term of the sequence $t_n = 7n + 2$ is 261? 1

(a) $n = 38$ **(b)** $n = 39$ **(c)** $n = 37$ **(d)** $n = 36$

10. A number that is the product of two different primes has exactly: 1

(a) 6 factors **(b)** 3 factors **(c)** 4 factors **(d)** 2 factors

11. The converse of the proposition “If a quadrilateral is a rhombus, then its diagonals bisect each other.” is: 1

(a) A rhombus has diagonals that bisect each other.

(b) If the diagonals of a quadrilateral do not bisect each other, then it is not a rhombus.

(c) If a quadrilateral is not a rhombus, then its diagonals do not bisect each other.

(d) If the diagonals of a quadrilateral bisect each other, then it is a rhombus.

12. Marks of 50, 80, 65 are combined with weights 5 : 3 : 2. The weighted average is: 1

(a) 61 **(b)** 64 **(c)** 63 **(d)** 62

13. A house used 200, 300, 100 and 400 units of electricity on lighting, cooling, kitchen and other appliances in 2010. In a 100% stacked bar for 2010, the length of the cooling segment is: 1

(a) 60% **(b)** 30% **(c)** 35% **(d)** 25%

14. The averages of 3 weeks of 5 days each are 12, 15, 18. The overall average is: 1

(a) 17 **(b)** 14 **(c)** 15 **(d)** 16

15. The improved gcd algorithm that keeps most-recent-common-divisor scans k from 2 up to min(54000, 81000). The number of values of k scanned is: 1

(a) 53998 **(b)** 54000 **(c)** 80999 **(d)** 53999

16. The number of times Step 3 of Euclid's subtraction algorithm is applied while computing gcd(50, 8) is: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 12

17. The median AD of $\triangle ABC$ has length 30 cm and G is the centroid. Then AG is: 1

(a) 10 cm **(b)** 15 cm **(c)** 28 cm **(d)** 20 cm

18. The slope of the line through the points (-3, -2) and (1, 4) is: 1

(a) -3/2 **(b)** 5/2 **(c)** 3/2 **(d)** 2/3

19. The y-intercept of the line 3x − 4y = 12 is: 1

(a) 4 **(b)** 3 **(c)** -3 **(d)** -4

20. Taking $\pi = \frac{22}{7}$, the curved surface area of a cylinder of radius 3.5 cm and height 12 cm is: 1

(a) 286 cm^2 **(b)** 462 cm^2 **(c)** 132 cm^2 **(d)** 264 cm^2

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. The midpoint of AB is $M(2, -1)$ and $A = (-4, 3)$. Find B. 2

22. Find two rational numbers between $\frac{1}{2}$ and $\frac{3}{4}$. 2

23. Factorise $x^2 - 5x + 6$ by splitting the middle term. 2

24. Why does an inscribed regular hexagon show that $\pi > 3$? 2

25. What does a ‘weight’ mean in a weighted average? 2

26. Name two real-life devices that use parallelogram properties. 2

27. Show that $x + 2y = 4$ and $3x + 6y = 12$ have the same solutions. 2

28. Find the volume of a pyramid with a square base of side 5 cm and a height of 12 cm. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. The points $(1, -8)$, $(-4, 7)$ and $(-7, -4)$ lie on a circle with centre at the origin. Find its radius and say whether $(-5, 6)$ lies inside, on or outside the circle. 3

30. A rational number in its lowest form has denominator $2^3 \times 5$. How many decimal places will its expansion have? Explain. 3

31. If $x + \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$ and $\left(x - \frac{1}{x}\right)^2$. 3

32. ABCD is a cyclic trapezium with $AB \parallel CD$. Show that its non-parallel sides AD and BC are equal. 3

33. **Case study:** A kite-shaped signboard ABCD has diagonals AC $= 24$ cm and BD $= 10$ cm meeting at right angles; E, F, G, H are the midpoints of AB, BC, CD, DA. 3
(i) Name EFGH.
(ii) Find its sides.
(iii) Find its perimeter.

34. Baudhāyana's method: find the side of a square whose area equals that of a $16\ \text{cm} \times 9\ \text{cm}$ rectangle. 3

35. Explain what happens to experimental probability as the number of trials increases, using the Law of Large Numbers. 3

OR

A card is drawn from a pack of 52. Find the probability of 3
(i) a black queen
(ii) the ace of spades
(iii) a number card from 2 to 10.

36. Write the converse of "If the sum of the digits of a number is a multiple of 3, then the number is divisible by 3." Is it true? 3

OR

Show with an example that "If two triangles have equal areas, then they are congruent" is false, and write its converse and its truth. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Explain 5
(i) why a proposition and its converse can both be true, using the statement about equal angles opposite equal sides in a triangle, and
(ii) the example of both being false.

OR

Case study: The statement "If a number is divisible by 6, then it is divisible by 3" is checked on the numbers 18, 24, 30. 5
(i) Verify.
(ii) Write the converse.
(iii) Test it on 9, 15, 21, 12 and find which of these are counterexamples.
(iv) What fraction of the tested numbers is divisible by 6?

38. (i) Find the mean depth of a pit (Brahmagupta's rule) if the depth is 2 m for a length of 3 m, 5 m for 4 m and 3 m for 3 m. 5
(ii) Explain why a simple average of the depths would be misleading.

OR

(i) The average daily rainfall in August, September and October is 12 mm, 6 mm and 2 mm. Write the expression for the combined average per day. 5
(ii) Evaluate it for 31, 30 and 31 days.

39. Prove that $(x+y+z)(x^2+y^2+z^2-xy-yz-zx) = x^3+y^3+z^3-3xyz$ and evaluate $1^3+2^3+(-3)^3$. 5

OR

(i) Prove $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$ using $(a+b)(a+b)^2$. 5

(ii) Deduce $(a-b)^3$ and use it to compute 97^3.

40. **Case study:** A student has ₹500 in her savings account and gets ₹150 every month. 5

(i) Write a linear expression for the amount after n months.

(ii) Find the amount after 6 months.

(iii) After how many months will she have ₹2,000?

(iv) Plot (describe) the graph and identify the slope.

OR

Case study: Reiaan's school sells notebooks at ₹x each. The cost for n notebooks including a one-time ₹20 bag fee is $20 + x\,n$. For 5 notebooks the bill is ₹70. 5

(i) Find x.

(ii) Write the cost function.

(iii) Find the cost of 12 notebooks.

(iv) How many notebooks can be bought for ₹170?

END OF QUESTION PAPER **Total: 80 Marks**

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